# Confinement-loss prediction in photonic crystal fiber SPR sensors: kernel surrogates versus generative data augmentation

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**Abstract**

Designing photonic crystal fiber (PCF) based surface plasmon resonance (SPR) sensors requires repeated full-vectorial finite element simulation, which makes iterative geometric optimisation prohibitively slow. Machine-learned surrogates are the standard remedy, and the prevailing strategy is to grow the scarce simulation set with a generative adversarial network before fitting a deep network. We show that this strategy is counterproductive for the kernel models that suit this data regime. On a 432-sample full-vectorial finite element dataset, a Bayesian-optimised support vector regressor reaches a mean squared error of 0.121 on the held-out geometry, against 0.314 for the published GAN-augmented deep network, a 61% reduction achieved without synthetic data. Adding the same synthetic samples degrades both kernel models: the support vector regressor rises to 0.197 and Gaussian process regression collapses entirely. We trace the collapse to its cause, exhibiting generated sample pairs that sit 0.011 apart in scaled feature space yet differ by 0.168 in log-scaled loss, driving the covariance matrix towards singularity. Shapley attribution identifies the outer cladding holes as the dominant control on confinement loss, and a Monte Carlo study identifies the lattice pitch as the dominant source of resonance instability under fabrication error, at 98.8% of the variance in peak position. Timing measurements show the generative route to be the most expensive stage of the pipeline while reducing accuracy. The resulting surrogate is interpretable, trains in under a minute, and needs 70% less simulation data than comparable neural approaches.

**Keywords:** Photonic crystal fiber, Surface plasmon resonance, Support vector regression, Explainable artificial intelligence, Bayesian optimisation, Surrogate modelling

## 1 Introduction

Photonic crystal fibers coated with a thin plasmonic film form the basis of highly sensitive surface plasmon resonance sensors for chemical and biological detection. A guided core mode is deliberately allowed to leak towards a metal surface; where the mode index phase-matches the surface plasmon polariton, the confinement loss spectrum develops a sharp resonance whose wavelength tracks the refractive index of the surrounding analyte. Locating and shaping that resonance is a geometric design problem over the cladding lattice, the plasmonic film and the analyte channel.

The established way to evaluate a candidate geometry is the full-vectorial finite element method (FV-FEM), which solves Maxwell's equations on the fiber cross section. FV-FEM is accurate but expensive, and a design study sweeping several geometric parameters across a wavelength grid quickly becomes a multi-day computation. This cost, rather than any modelling difficulty, is what limits practical PCF-SPR design.

Machine-learned surrogates address the bottleneck by learning the map from geometry and wavelength to optical response, so that a trained model replaces the solver during the sweep. Chugh et al. [3] first applied artificial neural networks to PCF property computation. Deep surrogates, however, need far more training data than FV-FEM can cheaply supply, and the field's response has been to manufacture the missing data: Zelaci et al. [23] trained a Wasserstein GAN with gradient penalty on their simulation set, used it to synthesise additional samples, and reported a mean squared error (MSE) of 0.314 for the resulting augmented network.

We argue that this sequence solves a problem that need not be created. The quantity being modelled is smooth in geometry apart from a sharp resonance peak, the sample count is small ($N < 500$), and the input space is six-dimensional. These are the conditions under which margin- and kernel-based regressors are expected to outperform deep networks, and recent PCF studies are consistent with this [11, 16]. If a kernel model is accurate enough on the original data, the augmentation step – and the generative model behind it – is unnecessary.

This paper makes that case quantitatively, and then shows the augmentation step is not only unnecessary but detrimental. The contributions are as follows.

1. A Bayesian-optimised support vector regression (SVR) surrogate attaining an MSE of 0.121 on an unseen geometry from the 432 original FV-FEM samples alone: a 61% error reduction against the published GAN-augmented deep network, and a 70% reduction in data requirement against the closest neural alternative [4].
2. An *augmentation paradox*: the same synthetic samples degrade SVR and destroy Gaussian process regression (GPR). A mechanism is supplied rather than an observation, exhibiting generated pairs that are geometrically near-identical yet carry materially different targets, and showing how these drive the GPR covariance matrix towards singularity.
3. A Shapley-value decomposition of the surrogate that attributes confinement loss primarily to the outer cladding hole diameters, connecting the regression to the physics of evanescent leakage rather than treating it as an opaque predictor.
4. A Monte Carlo tolerance analysis ranking the geometric parameters by their contribution to resonance instability under fabrication error, and a measured comparison of the training and inference cost of every model considered.

A preliminary version of this work appeared in the proceedings of the 34th IEEE Signal Processing and Communications Applications Conference [8]. The present article extends it with a per-geometry cross-validation of every model, the mechanistic account and empirical evidence for the GPR collapse, the tolerance study of Sect. 5.4, and an expanded treatment of the optimisation and evaluation protocol. All figures and tables have been produced anew for this article.

Section 2 surveys related surrogate work. Section 3 describes the sensor and the FV-FEM model that generates the data. Section 4 sets out the surrogate methodology. Section 5 reports accuracy, the augmentation paradox, feature attribution, tolerance and spectral validation. Section 6 discusses limitations, and Sect. 7 concludes.

## 2 Related work

***Neural surrogates for fiber optics.***

Chugh et al. [3] established the template: sample the geometric design space with FV-FEM, fit a feed-forward network, and use it for fast sweeps. The approach is accurate where data is plentiful, and its weakness is precisely that requirement. Ehyaee et al. [4] pushed accuracy to an MSE of 0.084 on a PCF-SPR sensor, but required 1,476 simulated samples to achieve it. The reported accuracy is not in question; the associated simulation budget is the cost of obtaining it.

### *Generative augmentation.*

Zelaci et al. [23] attacked the data requirement directly, training a WGAN-GP [7] on their FV-FEM set and augmenting the training data with its output. Reported MSE improved from 0.894 to 0.314. This is the baseline against which the present work is positioned. The improvement is measured relative to a deep network that was itself limited by sample count, and the comparison does not establish whether a model better suited to small datasets would have required the synthetic samples at all.

### *Design optimisation without deep networks.*

A parallel line of work optimises PCF-SPR geometry without a deep surrogate at all. Kaziz et al. [13] combined a Taguchi design of experiments with machine learning and a genetic algorithm; Romeiro et al. [19] studied geometric optimisation of the cladding lattice directly. Goswami et al. [6] classified guided modes of a PCF-SPR sensor, and Fasseaux et al. [5] learned refractive-index dynamics for plasmonic fiber biosensors from measured rather than simulated spectra. These share with the present work the premise that the sample budget, not model capacity, is the binding constraint.

### *Kernel methods on small optical datasets.*

Kalyoncu et al. [12] used $k$-nearest-neighbour regression for the loss characteristics of bent PCF-SPR sensors. Kaium and Mollah [11] applied GPR to PCF property estimation, and Opoku et al. [16] benchmarked ten regressors on a simulated PCF-SPR sensor, with kernel and ensemble methods leading. Applications of the same family of sensors continue to broaden [1]. These results motivate our hypothesis but stop short of testing it against a generative-augmentation baseline on identical data, which is what we do here.

### *Interpretability.*

Khatun and Islam [14] applied Shapley additive explanations [15] to a PCF-SPR design, isolating gold film thickness as a driver of sensitivity, and Akter and Abdullah [2] argued for deliberately interpretable models in plasmonic sensor optimisation. We extend that style of analysis to a multi-family cladding lattice, where the question is not the thickness of one layer but which of several hole families controls leakage.

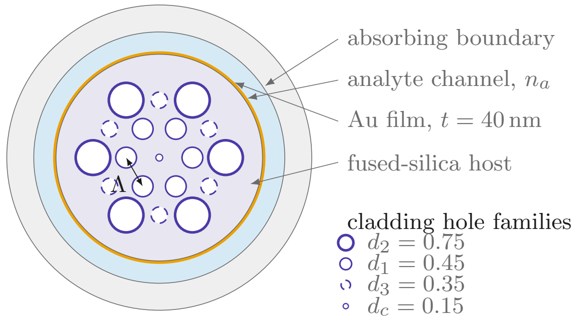


**Fig. 1** Cross section of the modelled PCF-SPR sensor. Three families of air holes sit on a hexagonal lattice of pitch $\Lambda$ in a fused-silica host, with a central defect hole of diameter $d_c$. Light leaking from the core couples to the surface plasmon supported by the gold film. Legend diameters are in µm and give the nominal design; the dataset sweeps them over the ranges in Table 1.

## 3 Sensor structure and numerical model

The dataset underlying this study was produced by FV-FEM analysis of the sensor cross section in Fig. 1, computed in COMSOL Multiphysics with perfectly matched layers terminating the computational domain. The structure comprises a fused-silica host, a hexagonal cladding lattice of air holes, a plasmonic gold film, and an outer analyte channel that acts as the sensing medium.

Nineteen air holes in three diameter families form the cladding, together with a small central defect hole. The gold film is 40 nm thick, with permittivity taken from the Johnson and Christy measurements [9]. The refractive index of the silica host follows the Sellmeier relation [18]

$$n^2(\lambda) = 1 + \sum_{i=1}^{3} \frac{B_i \lambda^2}{\lambda^2 - C_i}, \tag{1}$$

with $B_i$, $C_i$ the standard Sellmeier coefficients for fused silica. The predicted quantity is the confinement loss, obtained from the imaginary part of the effective index as [23]

$$L_c = \frac{40\pi}{\ln(10)\,\lambda}\,\mathrm{Im}(n_{\mathrm{eff}}) \times 10^4 \quad [\mathrm{dB/cm}]. \tag{2}$$

**Table 1** Sampled design space. Nine geometric configurations, each evaluated for three analytes over a 16-point wavelength grid, give $9 \times 3 \times 16 = 432$ FV-FEM samples.

| Variable | Role | Sampled values |
|---|---|---|
| $n_a$ | analyte index | 1.33, 1.34, 1.35 |
| $\lambda$ | wavelength | 16-point grid |
| $\Lambda$ | lattice pitch | 0.15, 0.20, 0.24 |
| $d_1$ | inner holes | 0.25, 0.45 |
| $d_2$ | outer holes | 0.55, 0.75 |
| $d_3$ | interstitial holes | 0.15, 0.35 |
| $d_c$ | central defect | 0.15 (fixed) |
| $L_c$ | target | $10^{-3}$–59.1 dB/cm |

Geometric values are the dataset's own column values. The $\lambda$ column is in units of 100 nm, so the 16-point grid spans 500–800 nm; the Sellmeier index of fused silica over that span, $n = 1.462$–$1.453$, matches the dataset's $\mathrm{Re}(n_{\mathrm{eff}})$ of 1.42–1.46.

$L_c$ peaks at the phase-matching condition, so the resonance region, where the function is least smooth, is what a surrogate must reproduce.

# 4 Surrogate modelling methodology

## 4.1 Dataset and preprocessing

We use the FV-FEM dataset of Zelaci et al. [23], so that the comparison against their GAN-augmented network is made on identical data. It contains 432 labelled samples over six input variables: analyte refractive index $n_a$, wavelength $\lambda$, lattice pitch $\Lambda$, and the three cladding hole diameters $d_1$, $d_2$, $d_3$. Table 1 summarises the sampled design space.

The target spans nearly five orders of magnitude and is dominated by the resonance peak, so we regress the log-scaled response $y = \log_{10}(L_c \times 10^8)$. This both compresses the dynamic range and makes squared error a sensible loss across the whole spectrum rather than one concentrated at the peak. Inputs are min-max scaled to $[0, 1]$, which matters for the radial basis function (RBF) kernel because an unscaled wavelength axis would otherwise dominate the distance metric.

## 4.2 Evaluation protocol

Random train-test splitting is inappropriate for this dataset. The 432 samples are not independent: they derive from nine geometric configurations, each contributing a full wavelength sweep. A random split places samples from the same geometry on both sides of the partition, so that the model interpolates within a spectrum it has already observed. The resulting error understates the quantity of interest, which is generalisation to a geometry that has not been simulated.

We therefore evaluate at the level of geometries, under two protocols:

- **Fixed partition.** The 7–1–1 split of [23]: configurations 1–7 for training, 8 for validation, 9 held out for test. This reproduces the published protocol exactly and is the basis for the headline comparison.
- **Leave-one-geometry-out (LOGO).** Each of the nine configurations serves in turn as the test set, the other eight as training data. This gives nine independent estimates and, unlike the fixed partition, reports the variance across geometries as well as the mean.

The fixed partition constitutes a single draw and may favour a model that happens to suit configuration 9. LOGO is the more informative of the two protocols; both are reported.

## 4.3 Support vector regression

SVR fits a function deviating from the observed targets by at most $\epsilon$ while remaining as flat as possible, penalising only the samples falling outside that tube [21]. Writing $C$ for the box constraint and $\xi_i$, $\xi_i^*$ for the slack variables, the primal problem minimises $\frac{1}{2}\|w\|^2 + C\sum_i(\xi_i + \xi_i^*)$ subject to the two $\epsilon$-tube constraints; the full formulation, its dual and the Karush-Kuhn-Tucker conditions are given in Online Resource 1, Sect. S1. The kernel is radial basis, $K(x_i, x_j) = \exp(-\gamma\|x_i - x_j\|^2)$.

The relevant property is that the fit is determined by the support vectors, the samples on or outside the tube boundary, rather than by all 432 points equally. Near the resonance, where the response turns sharply, this concentrates model capacity where the function is least smooth, which accounts for the method's tolerance of a small sample count.

### 4.4 Gaussian process regression

GPR places a Gaussian process prior $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$ over functions, so that the posterior mean and covariance at a test input $x_*$ are [17]

$$\bar{f}_* = K_*^\top \left[K + \sigma_n^2 I\right]^{-1} y, \tag{3}$$

$$\mathrm{cov}(f_*) = K_{**} - K_*^\top \left[K + \sigma_n^2 I\right]^{-1} K_*, \tag{4}$$

with $K$ the training covariance, $K_*$ the train-test covariance and $\sigma_n^2$ the noise variance. We use an RBF kernel with an additive white noise term, which regularises the inversion in (3) and (4). GPR offers calibrated uncertainty, which SVR does not, and that makes its behaviour under augmentation in Sect. 5.2 diagnostic: the same smoothness assumption that yields the uncertainty estimate is what the synthetic data violates.

### 4.5 Hyperparameter optimisation

$C$, $\gamma$ and $\epsilon$ interact, and grid search over three coupled continuous parameters wastes most of its budget in uninformative regions. We instead use Gaussian-process Bayesian optimisation [22] with expected improvement, 32 evaluations, and log-uniform priors over $C \in [1, 2000]$, $\gamma \in [10^{-4}, 1]$ and $\epsilon \in [10^{-4}, 10^{-1}]$, scoring each candidate by three-fold cross-validated MSE on the training partition only. The selected configuration and the full search space are tabulated in Online Resource 1, Sect. S2.

## 5 Results

### 5.1 Predictive accuracy

Table 2 reports every model under both protocols. On the fixed partition the optimised SVR reaches an MSE of 0.121, against 0.314 for the GAN-augmented deep network of [23] and 0.894 for the unaugmented network – a 61% error reduction relative to the augmented baseline, obtained without generating a single synthetic sample. GPR reaches 0.185, also well ahead of both neural baselines.

The LOGO protocol matters more, and it separates the models further. Averaged over the nine held-out geometries the SVR attains an MSE of 0.246 with a standard deviation of 0.226, against 0.825 ± 0.609 for the augmented benchmark. The gap in the standard deviation is the more useful number for a design tool: a surrogate that is accurate on average but occasionally wrong by a wide margin on an unfamiliar geometry cannot be trusted to drive an optimisation loop. Per-fold results are given in Online Resource 1, Sect. S3.

The comparison with Ehyaee et al. [4] requires careful statement. Their ML-enhanced ANN attains an MSE of 0.084, lower than the 0.121 reported here, but does so using 1,476 training samples against 432. Since each training sample is an FV-FEM solution, this difference dominates the cost of the entire pipeline, and a 70% reduction in that cost for a 0.037 increase in MSE constitutes a favourable trade-off in a design context.

The mechanism underlying this result is the one anticipated in Sect. 4.3. A deep network fits a global function and requires sufficient samples to constrain it throughout the input domain, a condition nine geometries do not meet. SVR instead concentrates representational capacity on the support vectors bounding the transition between the high- and low-loss regions, and extrapolates from that boundary rather than from a globally interpolated surface.

### 5.2 The augmentation paradox

The augmentation strategy of [23] was next applied to the kernel models, generating synthetic samples with the same WGAN-GP configuration and adding them to the training partition. Since synthetic data is known to benefit models limited by sample count, the question addressed here is whether it also benefits models that are not so limited.

Table 3 shows the support vector regressor degrading from 0.121 to 0.197 and Gaussian process regression failing outright, and the direction of change is consistent across the leave-one-geometry-out folds (Online Resource 1, Sect. S3). Augmentation is therefore not introducing noise that averages out over folds; it is displacing the fit. On this evidence, WGAN-GP augmentation does not improve predictive accuracy in either case.

The GPR failure is the informative case, because it is not a gradual degradation but a numerical breakdown, and its cause is identifiable. A GAN is a probabilistic sampler with no constraint tying it to Maxwell's equations, so

**Table 2** Accuracy of each surrogate under both evaluation protocols. Rows are metrics and columns are models, so that the fixed-partition result and the distribution over the nine leave-one-geometry-out folds can be read against each other for a given model. Lower is better throughout; dashes mark figures not reported in the source work.

| | Published neural baselines | | This work | | |
|---|---|---|---|---|---|
| Metric | Deep net [23] | GAN-augmented [23] | **SVR** | GPR | ML-enhanced ANN [4] |
| Fixed-partition MSE | 0.894 | 0.314 | **0.121** | 0.185 | 0.084 |
| LOGO mean MSE | — | 0.825 | **0.246** | — | — |
| LOGO standard deviation | — | 0.609 | **0.226** | — | — |
| Training samples | 432 | 432 + synth. | **432** | 432 | 1,476 |
| Synthetic data | no | yes | **no** | no | no |

**Table 3** Effect of adding GAN-generated samples to the training partition, split out here from the accuracy comparison because it measures a different thing: not which model is best, but what augmentation does to a model that did not need it.

| Model | Real only | Augmented | Change |
|---|---|---|---|
| SVR | 0.121 | 0.197 | +63% |
| GPR | 0.185 | 23.844 | collapse |
| GPR (LOGO) | — | 52.86 | collapse |
| Deep network [23] | 0.894 | 0.314 | −65% |

Fixed-partition MSE except where marked. The GPR leave-one-geometry-out mean is carried from [8]; the generator is retrained per fold, so fold-level values scatter widely (32.7–76.1 in a partial re-run) and the mean is the stable quantity. The neural row is reproduced from [23] for contrast: the same synthetic samples that help a data-starved network harm both kernel models.

**Table 4** The three most contradictory nearest-neighbour pairs among 50,000 GAN-generated samples, ranked by target discrepancy per unit of feature distance. Quantities run down the rows and pairs across the columns; each column is one pair of generated samples that are near-identical in the input space but carry incompatible targets.

| Quantity | Pair A | Pair B | Pair C |
|---|---|---|---|
| Feature distance | 0.0111 | 0.0089 | 0.0088 |
| Target 1 ($\log_{10}$) | 4.4466 | 4.3722 | 4.3280 |
| Target 2 ($\log_{10}$) | 4.6144 | 4.4958 | 4.4347 |
| Target discrepancy | 0.1678 | 0.1236 | 0.1067 |
| Discrepancy/distance | 15.09 | 13.81 | 12.06 |

Feature distance is Euclidean in the min-max scaled seven-dimensional space the generator was trained on; targets are the $\log_{10}$-scaled confinement loss.

nothing prevents it from emitting two samples that are geometrically almost identical while carrying materially different losses. Such a pair is physically impossible – the same geometry at the same wavelength has one confinement loss – but it is entirely consistent with a generator that has learned the marginal distributions without the physics.

To confirm this we generated 50,000 samples, found each one's nearest neighbour in the scaled feature space, and ranked the resulting pairs by the ratio of target discrepancy to feature distance. Table 4 lists the three worst. The leading pair sits 0.011 apart in a space scaled to the unit cube, yet its targets differ by 0.168 in log-scaled loss, a 45% discrepancy in physical units. The search protocol is detailed in Online Resource 1, Sect. S4.

For GPR the consequence is numerical failure rather than degradation. The posterior in (3) requires inverting $[K + \sigma_n^2 I]$. Two inputs at distance 0.011 produce near-identical rows in $K$, so the matrix approaches singularity; the white-noise term $\sigma_n^2$ is what normally absorbs this, but it is calibrated to the noise level of the real data and is far too small to absorb a 0.168 jump. The inverse is then dominated by the ill-conditioned directions and the predictive variance explodes. SVR survives because its $\epsilon$-insensitive loss simply admits both contradictory points as bounded support vectors: the fitted function is degraded, but the optimisation remains well posed.

Generative augmentation therefore exchanges a data-quantity limitation for a data-consistency one. For a model whose accuracy is limited by sample count this exchange may be favourable; for a model that is not so limited it is unambiguously detrimental.

This is a physical instance of a failure the wider machine-learning literature has begun to document. Shumailov et al. [20] showed that training

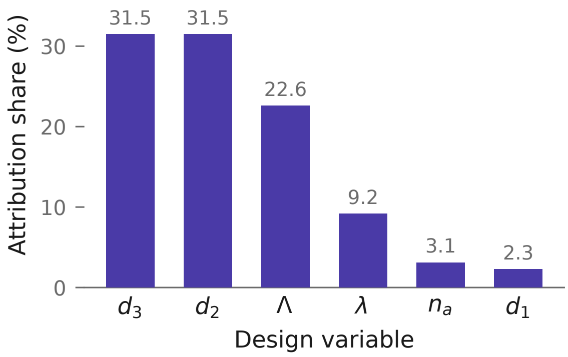


**Fig. 2** Global Shapley attribution for the trained SVR, expressed as each variable's share of total attributed magnitude. The outer cladding families $d_2$ and $d_3$ jointly account for 63% of the attribution; the innermost holes $d_1$ are nearly inert.

on model-generated data drives progressive degradation, and Juwara et al. [10] found synthetic augmentation beneficial only under conditions that must be checked rather than assumed. Our setting differs from the recursive loop those studies analyse, in that augmentation is performed once from a generator trained on real data. A single generation affords no opportunity for error to compound, so the degradation observed here must have another source. That source is the absence of a physical constraint: the generator is free to violate the single-valuedness that the underlying Maxwell problem guarantees.

### 5.3 What the surrogate learned

A surrogate that is accurate but opaque is of limited use in design, where the question is not only what the loss will be but which parameter to change. We applied Shapley additive explanations [15] to the trained SVR, which decomposes each prediction into additive per-feature contributions.

Figure 2 gives the resulting global ranking. The outer cladding diameters $d_2$ and $d_3$ dominate, together accounting for 63% of total attribution, followed by the lattice pitch $\Lambda$ at 23%. Wavelength contributes 9%, and the innermost holes $d_1$ barely register at 2%.

This ordering is physically consistent, and reading it against the geometry of Fig. 1 explains why. Confinement loss is set by how much of the core field reaches the gold surface, and the outer hole families are what that field must traverse to get there. Enlarging $d_2$ or $d_3$ raises the index contrast of the outer cladding, tightening confinement and suppressing leakage. The innermost holes sit inside the field maximum, where they shape the mode profile but do very little to gate its escape – hence their negligible attribution. That the model recovers this ordering from data alone, without being told the physics, is evidence it has learned the leakage mechanism rather than a correlation.

### 5.4 Manufacturing tolerance

Fiber drawing does not reproduce a design exactly, so a surrogate is useful for design only if it can also identify which deviations matter. The geometric inputs were perturbed with Gaussian noise at 1%, 3% and 5% relative standard deviation, 100 Monte Carlo trials were run at each level, and the induced scatter in the predicted resonance wavelength was recorded. The experiment was then repeated perturbing one variable at a time in order to attribute that scatter. Peak jitter grows from 2.3 nm at 1% tolerance to 10.1 nm at 5%; the complete tables are given in Online Resource 1, Sect. S6.

The pitch $\Lambda$ dominates, contributing 5.49 nm of jitter and 98.8% of the variance in peak position, while the three hole diameters together contribute under 0.7 nm. This establishes a fabrication priority: pitch uniformity along the drawn fiber governs whether a device matches its calibration, whereas tolerance on the cladding holes may be relaxed without appreciable spectral cost.

Two points qualify these values. First, they supersede the corresponding figures in [8], which were computed at a baseline geometry lying outside the sampled design space of Table 1; the surrogate was there extrapolating, and the resulting peak search returned grid endpoints in roughly half of all trials. The values above are obtained at a sampled geometry with a boundary-guarded peak search, under which no trial is rejected at any tolerance level. Second, the analysis perturbs the surrogate inputs only, and therefore propagates geometric tolerance but not the predictive error of the surrogate itself.

### 5.5 Computational cost

The argument for a surrogate rests on cost, so the cost was measured rather than asserted. Every model was timed on the same machine (AMD

Ryzen 5 5500U, 12 threads, CPU only), the same 432-sample dataset and the same software stack. The measurements are plotted in Online Resource 1, Sect. S8.

Three observations follow. First, fitting the SVR at fixed hyperparameters takes 39 ms, and the full 32-evaluation Bayesian search 31.1 s, so the entire surrogate is produced in well under a minute. Second, inference costs 14 µs per design point against 154 µs for the ANN, an order of magnitude that matters when a sweep evaluates $10^5$ candidate geometries. Third, and most relevant to Sect. 5.2, training the WGAN-GP generator alone takes 847 s, some four orders of magnitude more than the SVR fit it is intended to support. The augmentation route therefore incurs by far the largest computational cost in the pipeline while, on this dataset, reducing accuracy. These figures exclude the FV-FEM solutions themselves, which dominate the total budget in every case and are common to all models.

### 5.6 Spectral validation

Aggregate error statistics can conceal a surrogate that is accurate on average yet incorrect in spectral shape. Sweeping the trained SVR across the full wavelength range at fixed geometry reproduces the expected behaviour: a single well-formed resonance shifting monotonically to longer wavelengths as the analyte index rises from 1.33 to 1.35, in near-uniform steps of 27.93 and 27.33 nm, giving a spectral sensitivity of 2763 nm/RIU. The phase-matching condition was never supplied to the model, so recovery of both the direction and the regularity of this shift indicates that the underlying coupling has been captured rather than the training spectra memorised. The predicted spectra are shown in the Online Supplement, Sect. S7.

## 6 Discussion and limitations

The results support a claim narrower than a general superiority of kernel methods over deep learning. Deep surrogates underperform here not by virtue of their depth, but because nine geometries do not adequately constrain a global function approximator; synthesising further data addresses the symptom rather than the cause. Where sufficient real simulation data is available the neural result of [4] is stronger; the regime addressed here is the more common one in which it is not.

Three limitations bound this work. First, the analysis rests entirely on FV-FEM output, so the surrogate inherits any error present in the numerical model, and experimental validation on a drawn fiber remains necessary. Fabrication effects outside the model, including gold surface roughness, material impurity and ellipticity of the drawn holes, will affect sensitivity and signal-to-noise ratio in ways that surrogate accuracy cannot anticipate.

Second, the surrogate is interpolative within the sampled design space of Table 1, and is not a substitute for FV-FEM outside those ranges. The LOGO protocol measures generalisation to an unseen geometry within the sampled envelope rather than beyond it.

Third, the augmentation paradox is demonstrated for a single generator architecture on a single dataset. The mechanism identified here, namely a generator subject to no physical constraint emitting inconsistent near-duplicate pairs, is not specific to WGAN-GP, and is expected to affect any unconstrained generative augmentation supplying a kernel model. Establishing this requires evaluation of generators constrained to respect the governing equations, which would provide the additional data volume without the accompanying inconsistency. A consistency check on generated samples, of the kind performed in Sect. 5.2, should in any case precede the use of synthetic data in this setting [10].

## 7 Conclusion

We have shown that a Bayesian-optimised support vector regressor predicts PCF-SPR confinement loss more accurately than a GAN-augmented deep network on identical data – a 61% error reduction – using only the 432 original FV-FEM samples and no synthetic data. Under leave-one-geometry-out validation the advantage holds in both mean and variance.

The augmentation step that the surrogate literature has converged on is, for this class of model, actively harmful. Adding GAN-generated samples raises SVR error by 63% and collapses GPR entirely, and we have traced the collapse to generated sample pairs that are geometrically near-identical yet carry incompatible targets, driving the covariance matrix towards singularity. Generative augmentation exchanges a shortage of data for an inconsistency in it; that exchange is only worth making for a model that cannot function without the volume.

Beyond accuracy, the surrogate yields design guidance: Shapley attribution identifies the outer cladding diameters as the dominant control on confinement loss, and Monte Carlo tolerance analysis identifies the lattice pitch as the dominant source of resonance instability. Together these indicate both the parameter to tune and the parameter to hold, at a training cost of tens of milliseconds and an evaluation cost of tens of microseconds.

**Supplementary information.** Online Resource 1 contains the SVR dual formulation and KKT conditions (Sect. S1), the Bayesian optimisation search space and selected hyperparameters (Sect. S2), per-fold leave-one-geometry-out results (Sect. S3), the WGAN-GP configuration and contradiction-search protocol (Sect. S4), a GPR kernel and regulariser ablation (Sect. S5), and the full Monte Carlo tolerance tables (Sect. S6).

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# Declarations

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**Data availability.** The dataset analysed in this study was generated in the course of [23] and is publicly available at https://github.com/Aimen-Zelaci/SPRPCF_ANN.

**Ethics approval and consent to participate, consent for publication, and materials availability.** Not applicable.

**Code availability.** Available from the corresponding author on reasonable request.

**Author contribution.** M.K.H. implemented the surrogate models, the augmentation experiments and the analyses, and drafted the manuscript. C.K. supervised the machine-learning methodology. A.Y. provided the photonic modelling and FV-FEM dataset. All authors revised and approved the manuscript.

**Prior publication.** A preliminary version was presented at the 34th IEEE Signal Processing and Communications Applications Conference (SIU 2026) [8]; Sect. 1 describes how this article extends it. All figures and tables are newly prepared.

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